

Initial Project Plan

EXOhSPEC Development: Empowering Precision with Closed-Loop Feedback Control

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1 Introduction

The Exoplanet High-Resolution Spectrograph (EXOhSPEC) is a pioneering project with a focus on experimental development. The goal of this research experimental project is to develop a closed-loop feedback control system using high-precision IDS [1] sensor and spectrograph. The sensor has an exceptional 10 MHz bandwidth and 1 picometer (pm) displacement resolution. This system will use the sensors' capabilities to adjust for environmental variables like temperature, pressure, humidity, refractive index and other elements that may affect the precision of the observations.

EXOhSPEC will undergo rigorous testing in various settings. These evaluations will include experiments in a controlled lab environment and observations of our Sun and neighboring stars. We'll also utilize automated telescopes locally and carry out tests in observatories located in the Canaries and Thailand. The design of EXOhSPEC, which strives to minimize the amount of optical components, is one of its most notable attributes. EXOhSPEC is a platform that holds great potential for advancing the field of high-resolution spectrographs, as it was designed to minimize the number of optical surfaces while maximizing efficiency and compactness. The project's long-term goals are noteworthy since they go beyond the boundaries of terrestrial exploration and include helping to construct highly sensitive space-based radial velocity equipment.

2 Scientific Background

The Exoplanet High-Resolution Spectrograph (EXOhSPEC) project is at the forefront of a rapidly advancing field in exoplanet research. Future missions [Newman et al., 2023], including HabEx and LUVOIR, are poised to catalog and study Earth-like planets orbiting nearby stars. A number of elements, most notably the frequency of Earth-like planets and our increasing understanding of the stars that host these planets, are critical to the success of these expeditions. For these missions, choosing target stars and maximising observation durations are critical tasks for radial velocity surveys, whether they are conducted on the ground or in space.

Recent innovations set the stage for EXOhSPEC's objectives. A compact in-plane echelle spectrograph features an actively controlled fiber-fed double-pass design, prioritizing efficiency and minimizing optical components. Real-time processing of arc lines and auto-guiding

software ensure exceptional precision in a laboratory setting [Jones et al., 2021]. Furthermore, problems with radial velocity spectrographs are addressed with a calibrator system built in cooperation with the European Southern Observatory, which is based on a Fabry-Perot interferometer. It focuses on wavelength calibration precision, a key element in detecting Earth-like planets. Impressive results show remarkable calibration stability over short and long periods, promising a future with exceptional calibration precision [Wildi et al., 2011]. To sum up, EX-OhSPEC aims to improve the accuracy and performance of exoplanet discovery by utilising recent developments in radial velocity spectrography. The overarching goal of this research is to improve our knowledge of planetary systems outside of our solar system.

3 Project Tasks and Timeline

The EXOhSPEC project is divided into distinct phases, each with a set of specific tasks.

Phase 1: Equipment Orientation and System Introduction (October)

In this phase, engage in thorough immersion in documentation and equipment familiarization, gaining an understanding of the system's components and operation. This will be followed by familiarization with the IDS sensor, including basic code execution and comprehension of its capabilities.

Phase 2: Closed-Loop Feedback System Development (November-December)

The second phase encompasses several crucial tasks. Build the foundational elements of the closed-loop system, using IDS data to control temperature through TEC 1091. This is complemented by the integration of the BME680 with Arduino to create a more efficient interface. A software template will be designed for interfacing with all devices and generating a data log. The refractive index correction, using BME680 data (Ciddor/Edlen), will be implemented to enhance accuracy. The heart of this phase is the development of a feedback control system utilizing IDS sensor data to make real-time adjustments. It will undergo two stages of implementation testing to assess its effectiveness.

Phase 3: System Integration and Optimization (January-February)

This phase focuses on the assembly of the integrated setup, bringing together the IDS sensor and spectrograph for preliminary testing. The fine-tuning of optical components will ensure that optical elements within the spectrograph are precisely calibrated for optimal light capture and signal quality. Extensive experiments will be conducted to validate the system's stability and sensitivity under varying conditions and light sources. The project also aims to advance the feedback mechanism, integrating pressure and beam control for enhanced performance. The overall performance, especially regarding stability and accuracy, will be rigorously evaluated.

Phase 4: Advanced Integration and Testing (March)

This phase takes testing to a more challenging level with environmental robustness testing, assessing the system's performance under varying real-world conditions. It also involves enhancing the sensor capabilities and configuring the interferometer for corrected distance measurements. Fine-tuning and calibration will ensure the system's precision is optimized. Extensive testing with various light sources, including the Sun, and stars, will validate the system's capability in real astronomical scenarios.

Phase 5: Documentation and Reporting (April)

In the final phase, compile findings, observations, and results into a comprehensive project report. Simultaneously, prepare materials and presentations for sharing the project's

outcomes with the scientific community.

A detailed, descriptive timeline of the project tasks, including start and end dates for each phase and task, has been attached in the form of a Gantt chart with the report.

4 Mitigating Possible Issues

In the course, of the EXOhSPEC project, it's vital to anticipate and navigate potential challenges effectively. As the sole researcher, I've taken steps to mitigate these issues with precision and care. Technical challenges are a given, but I'll maintain open communication with my mentors and advisors, seeking their guidance and expertise when necessary. Adhering diligently to the project's timeline will enable me to manage time efficiently, while regular assessments of resource availability and careful planning will act as safeguards against potential resource limitations.

Moreover, I'll implement rigorous data validation processes and prioritize equipment maintenance to address data anomalies and equipment failures head-on. To contend with external factors such as unpredictable weather, I'll schedule observations thoughtfully and reschedule when necessary. Precision in calibration procedures is paramount in avoiding potential experiment calibration issues. Addressing communication and reporting delays will involve proactive updates to advisors and secure data handling for the sake of confidentiality. Lastly, I'll prepare for unforeseen circumstances with contingency plans, including having backup personnel on hand where feasible.

In conclusion, by taking a proactive, vigilant, and adaptable approach, I am committed to ensuring the smooth progress of the EXOhSPEC project, even when faced with challenges. It is through these measures that we can remain on track to achieve the project's objectives while navigating potential obstacles with resilience and expertise.

References

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- HRA Jones, WE Martin, Guillem Anglada-Escudé, Ronny Errmann, DA Campbell, Carl Baker, C Boonsri, and P Choochalerms. A small actively controlled high-resolution spectrograph based on off-the-shelf components. *Publications of the Astronomical Society of the Pacific*, 133(1020):025001, 2021.
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